

FLOODSENSE

Expert System-Based Flood Susceptibility Android Application

Feature Specifications & Complete System Flow

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1. Document Overview

This document details all features of the FloodSense Android application, including the revised features and newly proposed enhancements developed in response to the thesis panel's feedback and the thesis adviser's guidance. It covers the complete user flow from app launch to every available interaction, the purpose of each feature, implementation notes, and feasibility rationale.

FloodSense was originally developed as a pre-event flood awareness tool that combined machine learning-based susceptibility classification with augmented reality visualization. Following revisions directed by the thesis adviser, the system has been substantially redesigned. The AR camera-based map has been replaced with a pre-rendered, scenario-driven susceptibility map; the machine learning component has been removed in favor of an Expert System approach grounded in knowledge formally sourced from the Bacoar Disaster Risk Reduction and Management Office (BDRRMO); and the development framework has been

migrated from Unity (which was chosen specifically for AR capabilities) to Flutter, which is better suited for a multi-screen, data-driven Android application.

In response to the panel's challenge — 'What happens next after users become aware?' — the Action Checklist feature has been elevated into a full Decision Support System (DSS) integrated with an Expert System. This system presents a guided decision tree informed by BDRRMO expert knowledge, followed by a conversational AI assistant trained on expert-provided data to answer pre-event flood awareness queries. Together, these revisions transition FloodSense from a passive awareness tool to an active, expert-guided flood preparedness system.

2. Complete System Flow

The following is the complete step-by-step flow a user experiences from opening the app for the first time to accessing every available feature in FloodSense.

2.1 First-Time Launch Flow

STEP 1 — App Launch

User opens FloodSense on their Android device.

STEP 2 — Onboarding & Disclaimer Screen

First-time users are shown the onboarding screen explaining the app's purpose, scope, and limitations. A disclaimer states that susceptibility levels are expert-assessed pre-event estimates, not real-time flood guarantees. Users also receive an introduction to the Expert System component. User taps 'I Understand' to proceed.

STEP 3 — Login / Registration Screen

User either logs in with existing username and password or registers a new account. Firebase Authentication handles credential management. New users provide a username, password, and their home barangay.

STEP 4 — Main Dashboard / Home Screen

After login, the user lands on the main screen with access to all core modules: Rainfall Input, Susceptibility Map, Results Summary, Expert System (DSS + AI Assistant), Account & Preferences, and the action features.

2.2 Core Feature Flow (Revised Features)

STEP 5 — PAGASA-Aligned Rainfall Input

User selects rainfall intensity (Light, Moderate, Heavy, Intense, Torrential) and duration (1, 3, 6, 12, 24 hours) from plain-language dropdown menus. The app internally maps these to mm/hr values. Saved preferences from a previous session are pre-filled automatically.

STEP 6 — Expert System Susceptibility Classification

The app retrieves the pre-computed susceptibility lookup table from Firebase Firestore based on the selected rainfall intensity and duration combination. Susceptibility classifications are derived from BDRRMO expert knowledge and validated field data — not from a machine learning model. No live inference occurs at runtime. Each barangay is assigned a susceptibility level: Low, Moderate, High, or Very High.

STEP 7 — Pre-Rendered Susceptibility Map View

User taps 'View Map'. A pre-rendered, color-coded barangay susceptibility map of Bacoor City is displayed on screen. The map updates automatically to reflect the currently selected rainfall scenario (intensity and duration). No camera activation is required. The map is not GPS-anchored; it is a full-city static visualization. Users can pinch-to-zoom to examine individual barangay labels. A color legend and disclaimer remain accessible during map view.

STEP 8 — Results Summary Screen

A scrollable list of all Bacoar City barangays with their color-coded susceptibility badges is displayed. This provides an at-a-glance overview without requiring the map interface.

2.3 Action Feature Flow (Post-Awareness)

After viewing the susceptibility map or the Results Summary, the following action features are presented to the user, transitioning the app from passive awareness to active, expert-guided preparedness.

STEP 9 — Check My Area (Safety Status Check)

User taps 'Check My Area'. A map of Bacoar City is displayed with a draggable pin marker. The user manually drags the pin to their preferred location — no GPS permission is required. The app cross-references the pin's coordinates against a GeoJSON boundary file to identify the barangay. A results screen displays the identified barangay name, the active rainfall scenario, the corresponding susceptibility level with its color indicator, and a plain-language safety status summary. The result updates automatically if the user changes the rainfall scenario.

STEP 10 — Expert System DSS: Decision Tree with Integrated AI Assistant

Based on the susceptibility level returned for the user's pinned location, a guided decision tree is activated. The Expert System presents a structured sequence of situational questions (e.g., 'Are you currently at home?', 'Do you have a go-bag prepared?') and branches to level-appropriate, expert-informed action recommendations sourced from BDRRMO. At every step of the decision tree, an integrated AI assistant — trained on BDRRMO flood awareness data — is available for the user to ask contextual questions such as 'What should I put in a go-bag?' or 'What does High susceptibility mean for my barangay?' The assistant responds with expert-grounded, locally relevant answers informed by the user's current scenario and pinned location.

STEP 11 — Expert System AI Assistant: Follow-Up Chat

After completing the decision tree, the user is offered access to the Expert System AI Assistant — a conversational interface trained on BDRRMO expert knowledge and pre-event flood preparedness data. Users may type open-ended questions (e.g., 'What should I bring if I

evacuate with an elderly parent?', 'Which roads are usually flooded in Talaba during heavy rain?') and receive contextually informed responses. The assistant is aware of the user's current pinned barangay and active rainfall scenario.

STEP 12 — Find Nearest Evacuation Center

Triggered automatically when susceptibility is Moderate, High, or Very High. A ranked list of BDRRMO-designated evacuation centers is displayed, sorted by straight-line distance from the user's pinned location using the Haversine formula. Each entry shows the center name, barangay, and estimated distance. User taps 'Get Directions' and is deep-linked to Google Maps for turn-by-turn navigation.

2.4 Supporting Features

Account & Preferences

Users can view and update their registered barangay and saved rainfall preferences. Only one set of rainfall preferences is saved per account. Login is username and password only; social login is not supported.

Onboarding (Repeat Access)

Users can revisit the onboarding instructions and disclaimer at any time from the main screen via 'View Instructions and Parameters'.

3. Consolidated App Flow Summary

The following table summarizes the complete end-to-end flow of FloodSense from first launch through all available features.

#	Step / Feature	What Happens
1	App Launch	User opens FloodSense on Android device.

2	Onboarding & Disclaimer	First-time users read app purpose, scope, and disclaimer. Tap 'I Understand.'
3	Login / Register	Existing users log in. New users register with username, password, and home barangay.
4	Main Dashboard	Home screen with access to all modules.
5	Rainfall Input	User selects PAGASA-aligned intensity and duration from dropdowns.
6	Expert System Classification	App retrieves pre-computed BDRRMO-derived susceptibility lookup from Firestore for selected rainfall scenario.
7	Susceptibility Map View (REVISED)	Pre-rendered color-coded barangay map displayed. Updates based on selected rainfall scenario. No camera or GPS required.
8	Results Summary	Scrollable barangay list with color-coded susceptibility badges.
9	Check My Area	Map with draggable pin. Barangay resolved from pin coordinates. Results shown with barangay, scenario, susceptibility level, and safety summary.
10	DSS Decision Tree + AI Assistant (NEW)	Expert System presents guided situational questions and branches to BDRRMO-informed action recommendations. Integrated AI assistant answers flood awareness questions at any step of the decision tree.
11	AI Assistant Chat (NEW)	Conversational assistant trained on BDRRMO expert knowledge answers follow-up pre-event preparedness queries.
12	Find Nearest Evacuation Center	Ranked list of BDRRMO evacuation centers sorted by distance. 'Get Directions' deep-links to Google Maps.
13	Account & Preferences	User manages registered barangay and saved rainfall preferences.
14	Revisit Onboarding	User can re-read instructions and disclaimer from main screen at any time.

4. Feature Specifications

Feature 1 — Pre-Rendered Flood Susceptibility Map (REVISED)

The core visualization feature of FloodSense. Displays a pre-rendered, color-coded, barangay-level flood susceptibility map of Bacoar City. The map updates to reflect the currently selected rainfall scenario (intensity and duration). No camera, AR engine, or GPS is required.

Aspect	Details
Status	Revised core feature — replaces AR camera-based map
Technology	Flutter (Dart); pre-rendered map images or flutter_map vector overlay; Firebase Firestore for scenario-lookup
Input	Pre-computed susceptibility lookup table retrieved from Firebase Firestore based on user-selected rainfall parameters
Output	Color-coded map of Bacoar City barangays: Low (green), Moderate (yellow), High (orange), Very High (red)
Map Basis	BDRRMO expert-validated susceptibility data per barangay per rainfall scenario
Interaction	Pinch-to-zoom to examine individual barangay labels and susceptibility colors
Legend	Visible within map view. Full metadata accessible via in-app info panel.
Disclaimer	Accessible from map view at all times. States susceptibility is an expert-derived estimate, not a real-time flood guarantee.
GPS Used	No. Map is a static city-wide visualization, not anchored to the user's outdoor position.
Camera Used	No. This feature does not use the device camera.

Feature 2 — PAGASA-Aligned Rainfall Input (Existing)

Users specify a rainfall scenario through plain-language dropdown menus aligned with PAGASA's public advisory classification. The app handles all internal mm/hr conversion transparently.

Aspect	Details
Status	Existing core feature — unchanged
Intensity Options	Light (<2.5 mm/hr), Moderate (2.5–7.5 mm/hr), Heavy (7.5–15 mm/hr), Intense (15–30 mm/hr), Torrential (>30 mm/hr)
Duration Options	1 hour, 3 hours, 6 hours, 12 hours, 24 hours
Design Rationale	Plain-language labels remove the need for technical meteorological knowledge. Validated by BDRRMO key informant.
Saved Preferences	One set of rainfall preferences saved per user account via Firebase Firestore.

Feature 3 — Expert System Susceptibility Classification (REVISED)

Flood susceptibility per barangay is derived from expert knowledge formally sourced from BDRRMO, incorporating historical flood records, elevation data, and rainfall parameters. Results are pre-computed and stored as static JSON lookup tables in Firestore. No machine learning model is used; no live inference occurs at runtime. The Expert System replaces the previously proposed ML classifier.

Aspect	Details
Status	Revised core feature — replaces ML classification model
Knowledge Source	BDRRMO (institutional expert partner): historical flood records, field-assessed exposure data, barangay susceptibility assessments
Supporting Data	CPDCO elevation and land-use maps; PAGASA rainfall classification tables
Output Classes	Low, Moderate, High, Very High — one classification per barangay per rainfall scenario
Runtime Behavior	App retrieves pre-computed JSON from Firestore. No ML server or model inference required at runtime.
Rationale	Expert-sourced factual data is the appropriate basis for susceptibility classification. BDRRMO, as the institutional

	partner, serves as the subject-matter expert. The thesis adviser confirmed that machine learning is more appropriate for learning behavioral patterns, not for encoding expert factual knowledge.
Known Limitation	Classifications reflect expert-assessed static environmental conditions and historical records. Does not account for real-time drainage blockages or river levels.

Feature 4 — Results Summary & Decision Support (Existing)

A scrollable list of all Bacoar City barangays with color-coded susceptibility badges, populated from the pre-computed Firestore lookup table. Provides an at-a-glance overview without requiring the map interface.

Aspect	Details
Status	Existing core feature — unchanged
Technology	Flutter ListView widget; Firebase Firestore for data retrieval
Data Source	Pre-computed Firestore JSON document keyed by selected rainfall intensity and duration
Current Limitation	No sorting, filtering, or search functionality. All barangays displayed in a flat list.

Feature 5 — Check My Area: Location-Based Safety Status (Existing)

After viewing the susceptibility map, the user taps 'Check My Area' and is presented with a map of Bacoar City containing a draggable pin marker. The user manually positions the pin on their preferred location — no GPS permission is required. A dedicated results screen shows the identified barangay, the active rainfall scenario, the susceptibility level with its color indicator, and a plain-language safety status summary.

Aspect	Details
Status	Existing feature — unchanged

Trigger	User taps 'Check My Area' after viewing the map or Results Summary
Technology	Interactive map with draggable pin + GeoJSON barangay boundary file + existing Firestore lookup table
User Interaction	A map of Bacoar City is displayed with a draggable pin. The user drags and drops the pin onto their location. No GPS permission is requested or required.
Location Detection	Pin coordinates upon release are cross-referenced against the GeoJSON barangay boundary file. All processing is done locally on the device.
Rainfall Scenario Linkage	Result is dynamically tied to the rainfall scenario currently selected by the user. Changing the scenario refreshes the result for the same pinned location automatically.
Data Privacy	Compliant with RA 10173. No coordinates are transmitted or stored — pin placement and result computation are performed locally within the session only.

Feature 6 — Expert System DSS: Decision Tree with Integrated AI Assistant (NEW)

Addresses the panel's concern that awareness alone is insufficient and that the system must actively guide the user's decision-making. The Expert System DSS presents a structured decision tree immediately after the safety status card, asking the user a series of situational questions and branching to level-appropriate, expert-informed action recommendations. The knowledge base is sourced from BDRRMO and encoded into the system's rule set.

Integrated within this same feature is an AI assistant trained on BDRRMO expert knowledge and pre-event flood awareness data. As the user moves through the decision tree, the AI assistant is available at every step to answer flood awareness questions in context — such as what specific actions mean, why they are recommended, or what the user should prioritize given their situation. The AI assistant does not replace the decision tree; rather, it augments it by providing conversational, expert-grounded explanations alongside the structured guidance.

Aspect	Details
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Status	NEW — elevated from static Action Checklist; DSS layer of the Expert System with integrated AI assistant
Trigger	Automatically displayed after the Safety Status card (Feature 5) is shown
Technology	Rule-based decision tree encoded from BDRRMO expert knowledge; LLM API (e.g., Anthropic Claude API) trained on BDRRMO flood awareness data; Firebase Functions as secure middleware; Flutter widgets for screen rendering; Firestore for knowledge base storage
Knowledge Source	BDRRMO expert knowledge: pre-event preparedness protocols, evacuation decision criteria, flood awareness guidelines, safety actions by susceptibility level
Decision Tree Flow	The system presents a structured sequence of situational questions (e.g., 'Are you currently at home?', 'Do you have a go-bag prepared?') and branches to level-appropriate action recommendations based on the user's answers and their susceptibility level.
AI Assistant Role	The AI assistant is embedded within the decision tree interface. At any step, the user may ask flood awareness questions (e.g., 'What should I put in a go-bag?', 'What does High susceptibility mean for my area?') and receive expert-grounded, context-aware answers. The assistant is trained on BDRRMO data — not general internet knowledge — ensuring responses are accurate and locally relevant.
AI Training Basis	BDRRMO expert knowledge: flood history per barangay, pre-event preparedness protocols, evacuation procedures, road conditions during flood events, community flood awareness guidelines
Context Awareness	The AI assistant receives the user's current susceptibility level, pinned barangay, and active rainfall scenario as context with each query, enabling relevant and specific responses.
Implementation Complexity	Medium-High. Decision tree logic requires structured knowledge encoding from BDRRMO data. AI assistant requires LLM API integration with a curated BDRRMO-grounded system prompt and Firebase Functions middleware.

DSS Output by Level	LOW: Monitor and prepare. MODERATE: Secure valuables, identify evacuation routes, prepare go-bag, inform family. HIGH: Proceed to evacuation if advised, avoid flood-prone roads, contact relatives, follow BDRRMO and PAGASA advisories. VERY HIGH: Evacuate immediately, do not cross flooded roads, follow all BDRRMO and PAGASA emergency directives without delay.
Internet Dependency	The AI assistant requires an internet connection for each query. If connectivity is unavailable, the decision tree continues to function fully using locally stored rules. A fallback message informs the user that the AI assistant is temporarily unavailable.
Data Privacy	Compliant with RA 10173. User queries are processed via secure Firebase Functions middleware. No personally identifiable information is transmitted — only the contextual scenario data (barangay name, susceptibility level, rainfall scenario).

Feature 7 — Expert System AI Assistant: Follow-Up Chat (NEW)

After completing the decision tree, the user is offered access to an AI conversational assistant trained on BDRRMO expert knowledge and pre-event flood preparedness data. The assistant allows users to ask open-ended follow-up questions and receive personalized, expert-grounded responses. The assistant is context-aware — informed by the user's current pinned barangay and active rainfall scenario.

Aspect	Details
Status	NEW — AI chat layer of the Expert System
Trigger	Offered after the user completes the DSS Decision Tree
Technology	Large Language Model (LLM) API (e.g., Anthropic Claude API) with a curated system prompt embedding BDRRMO expert knowledge; Firebase Functions as secure middleware; Flutter chat widget for the conversational interface

Knowledge Basis	BDRRMO expert knowledge: flood history per barangay, evacuation protocols, preparedness guidelines, road conditions during flood events
Context Awareness	Assistant receives the user's pinned barangay name, active rainfall scenario (intensity and duration), and susceptibility level as context with each query
Example Queries	'What should I bring if I evacuate with an elderly parent?' 'Which roads in Talaba are usually flooded during heavy rain?' 'Is it safe to travel from Zapote to Molino right now under this scenario?'
Internet Dependency	Requires internet connection for each AI query. Behavior during low connectivity is handled gracefully with a fallback message directing users to the static DSS decision tree.
Cost Consideration	LLM APIs operate on a pay-per-use model. For thesis pilot testing (20–50 users), cost is estimated to be minimal. Full deployment cost to be evaluated post-thesis.
Data Privacy	Compliant with RA 10173. User queries are processed via secure Firebase Functions middleware. No personally identifiable information is transmitted with AI queries beyond the contextual scenario data.

Feature 8 — Find Nearest Evacuation Center (Existing)

Triggered automatically when the user's barangay is classified as Moderate, High, or Very High susceptibility. A ranked list of BDRRMO-designated evacuation centers is displayed, sorted by straight-line distance from the user's pinned location using the Haversine formula. The user taps 'Get Directions' to be deep-linked into Google Maps.

Aspect	Details
Status	Existing feature — unchanged
Trigger	Automatically offered after Safety Status card shows Moderate, High, or Very High result

Technology	Static evacuation center JSON stored in Firestore + Haversine formula for distance calculation + Google Maps URI deep-link
Evacuation Center Data	Sourced from BDRRMO as part of existing data-sharing collaboration. One-time data entry of center names, barangays, and GPS coordinates into Firestore.
Distance Calculation	Haversine formula computed on the device using the pin's dropped coordinates and each center's stored coordinates. No routing API or paid service required.
Display	Ranked list of nearest evacuation centers showing: center name, barangay, estimated straight-line distance from user.
Directions	Each center entry has a 'Get Directions' button that deep-links into Google Maps via geo: URI. All routing is handled by Google Maps.
Feasibility	High. No paid API required. Google Maps deep-link is a standard Android feature. Evacuation center list is a one-time static dataset obtainable from BDRRMO.

5. Proposed Revised Title

In light of the revised system architecture — replacing machine learning with an Expert System grounded in BDRRMO institutional knowledge — the thesis title has been updated to accurately reflect the nature of the system's intelligence and its expert knowledge source. The name 'FloodSense' is retained.

FloodSense: An Expert System-Based Flood Susceptibility Android Application for Pre-Event Flood Preparedness in Bacoar City

This title reflects: (1) the Expert System as the primary intelligent component, replacing the previously proposed ML classifier; (2) BDRRMO as the institutional expert knowledge source; (3) the system's expanded scope — from susceptibility awareness to active pre-event flood preparedness guidance; and (4) the geographic scope limited to Bacoar City.

6. Revised Technology Stack

The following reflects the updated technology stack following the removal of AR (Vuforia) and the transition from Unity to Flutter as the primary Android development framework.

Aspect	Details
Layer	Technology & Justification
Android App (All Screens)	Flutter (Dart) — purpose-built for multi-screen Android apps; cleaner and more efficient than Unity for non-AR applications; replaces Unity + uGUI
Map Display	flutter_map (open-source) or Google Maps Flutter plugin — supports pre-rendered overlays, color-coded barangay zones, and draggable pin markers; replaces Vuforia ground-plane AR
Backend / Database	Firebase (Firestore, Auth, Functions) — real-time sync, built-in authentication, cloud storage; unchanged from original design
Expert System Knowledge Base	Firebase Firestore — pre-computed susceptibility lookup tables and DSS decision tree rules encoded from BDRRMO expert data; replaces ML model and scikit-learn pipeline
AI Assistant	LLM API (e.g., Anthropic Claude API) with BDRRMO-grounded system prompt; Firebase Functions as secure middleware; Flutter chat widget for UI
Distance Calculation	Haversine formula (on-device, Dart implementation) — calculates straight-line distance from pin to evacuation centers; no routing API required
Navigation (Directions)	Google Maps URI deep-link — passes destination coordinates to installed Google Maps app; all routing handled by Google Maps
GeoJSON Boundary Lookup	Static GeoJSON file bundled with the app — used to resolve pin coordinates to barangay names; processed on-device
Elevation / Spatial Data	CPDCO (via data-sharing agreement) — topographic and land-use features for expert knowledge encoding
Historical Flood Data	BDRRMO (via data-sharing agreement) — ground-truth flood occurrence data per barangay; evacuation center list

Rainfall Reference	PAGASA Advisory Classification — basis for user-facing rainfall intensity presets; public domain
UI/UX Prototyping	Figma (free tier) — Android interface design and prototyping prior to Flutter development

7. New Features Feasibility Summary

The following table summarizes the feasibility of the revised and new features. All features operate within the existing Firebase free tier or minimal-cost API tiers and require no additional paid services beyond the LLM API used for the Expert System AI Assistant.

Feature	Complexity	Est. Dev Time	Key Dependency
Pre-Rendered Susceptibility Map	Low	1 week	Pre-rendered map images or flutter_map vector overlay + existing Firestore lookup; no camera or AR engine required
Expert System Classification	Low	1 week	BDRRMO expert knowledge encoded as pre-computed JSON lookup tables in Firestore; no ML training pipeline required
DSS Decision Tree + AI Assistant	Medium-High	2 weeks	Structured rule set encoded from BDRRMO knowledge; LLM API with BDRRMO-grounded system prompt; Firebase Functions middleware; Flutter widgets for rendering
AI Assistant Chat	Medium	1.5 weeks	LLM API (e.g., Anthropic Claude API) + Firebase Functions middleware + Flutter chat widget
Check My Area	Low-Medium	1 week	flutter_map or Google Maps Flutter plugin with draggable marker + GeoJSON boundary file + existing Firestore lookup

Find Evacuation Center	Nearest	Low- Medium	1 week	BDRRMO evacuation center list + Haversine formula + Google Maps deep-link
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Combined estimated development time for all revised and new features is approximately 7–8 weeks, which is well within a 10-month development timeline. The removal of the ML training pipeline, AR engine (Vuforia), and game engine framework (Unity) significantly reduces technical complexity compared to the original design. Flutter's first-class Firebase integration, built-in map plugins, and Android-optimized widget system make it a more appropriate and efficient framework for the revised FloodSense architecture.